

Analysis of edX Courses: Certification Rates Across Subjects

INTRODUCTION

edX, a platform developed by **MIT** and **Harvard**, provides access to **online courses** for millions of learners worldwide. However, not all courses achieve the same level of completion. This project investigates **how certification rates vary across different subjects** and institutions to better understand learner behavior and course effectiveness.



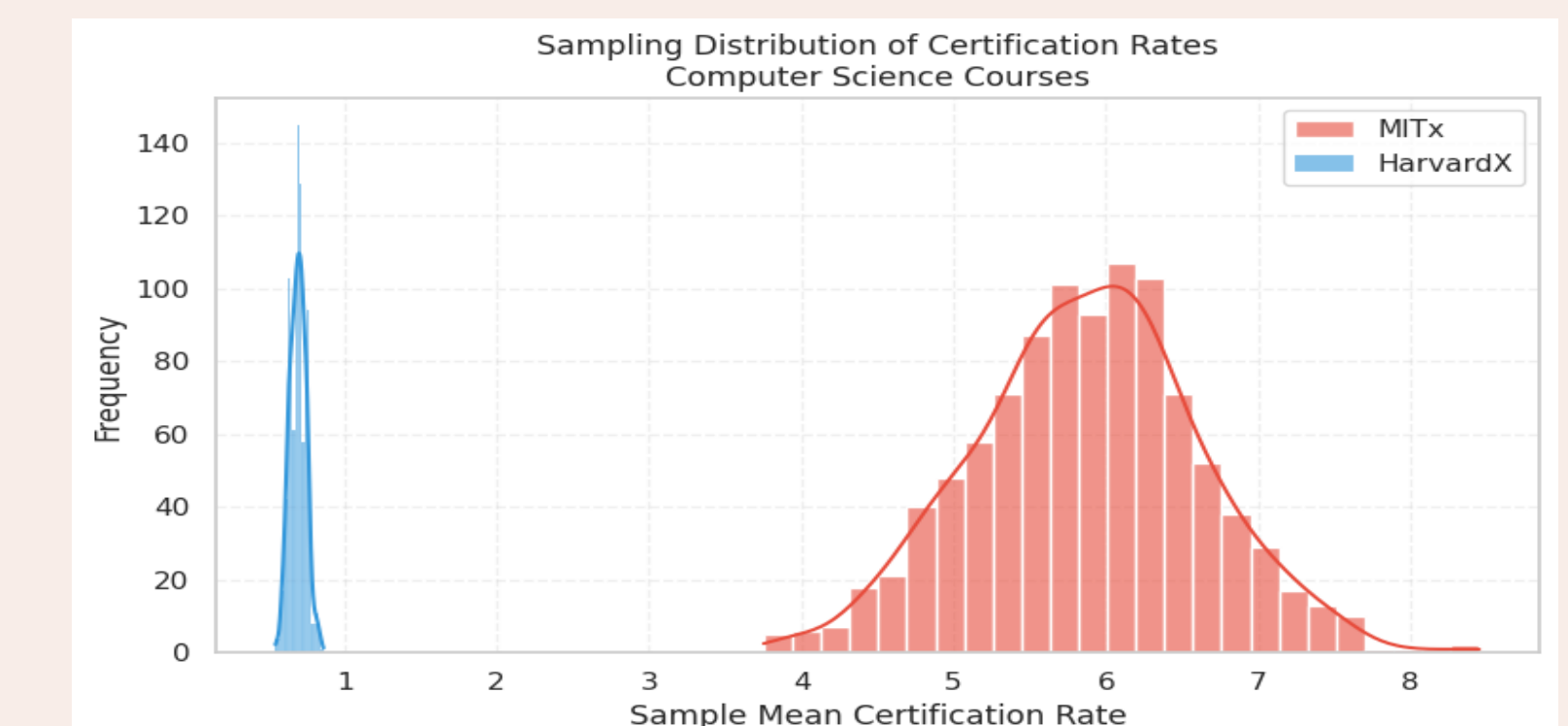
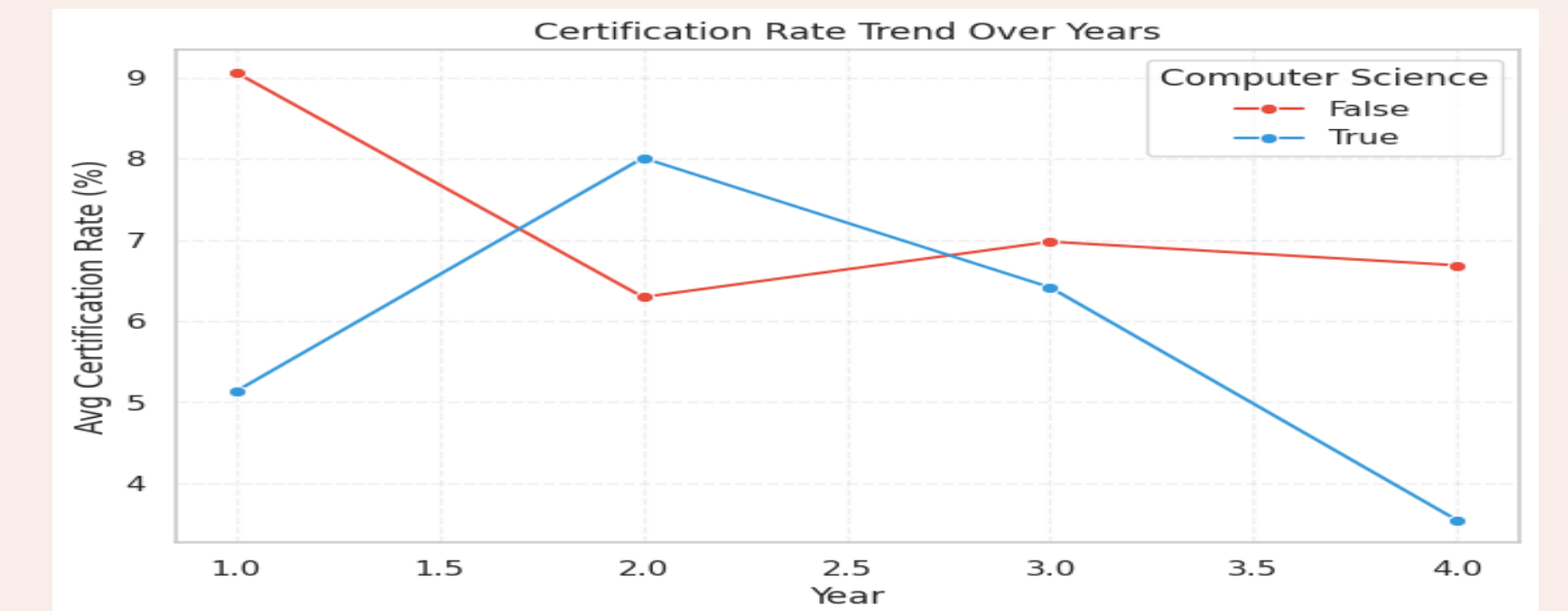
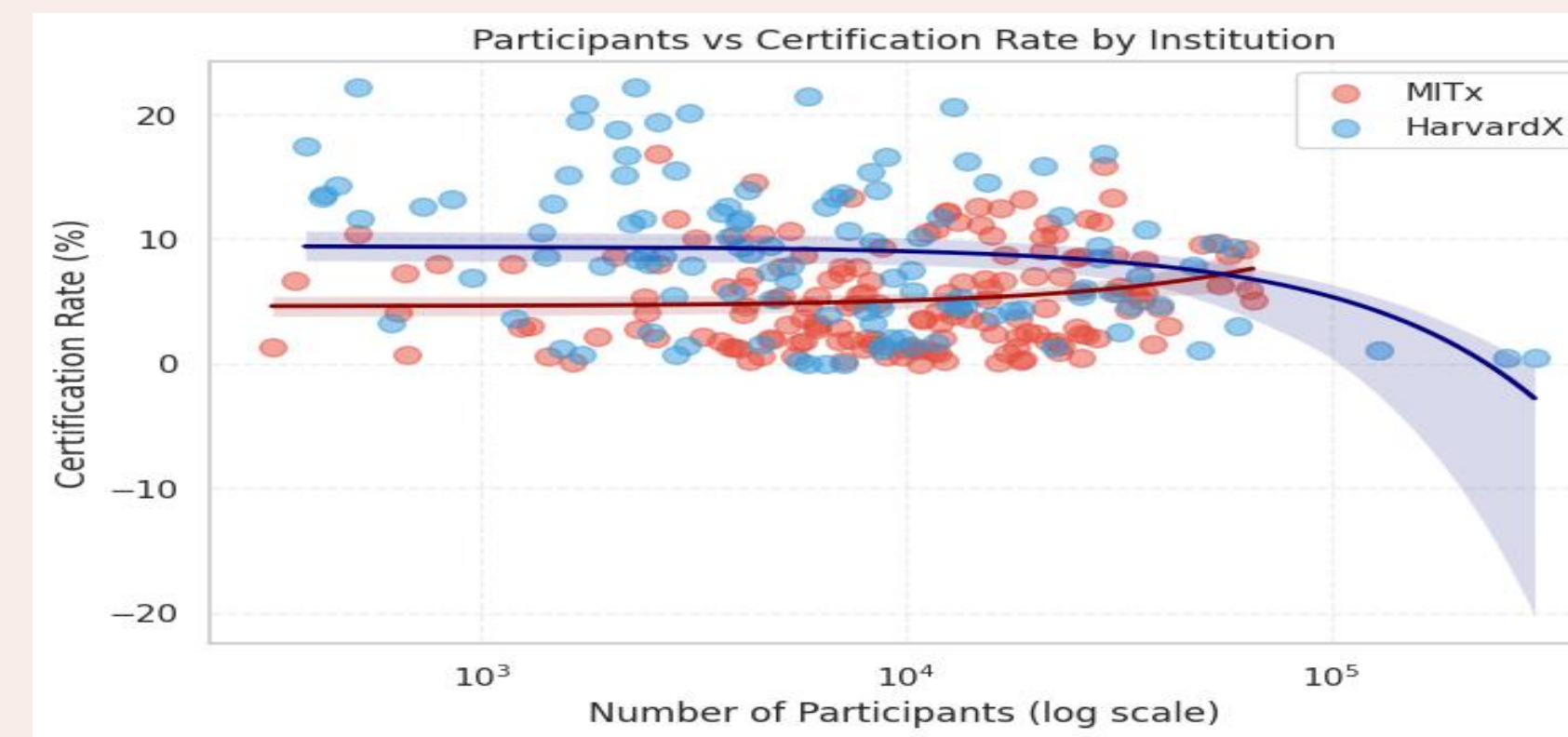
Research Question: Do certification rates differ significantly across different course subject areas?

Why It Matters:

Identifying these differences helps improve course design, increase student engagement, and enhance overall learning outcomes.

Analysis

Certification rates are generally low across most courses, with noticeable variation between subjects. Some fields achieve higher completion rates, while others remain lower, showing that subject area and course characteristics influence certification outcomes.



Hypothesis

Certification rates in Computer Science courses may vary between institutions (MIT and Harvard).

- H_0 : No significant difference in certification rates between MIT and Harvard (Computer Science)
- H_1 : There is a significant difference in certification rates between MIT and Harvard courses in Computer Science.

Hypothesis

Do certification rates differ significantly across different course subject areas?

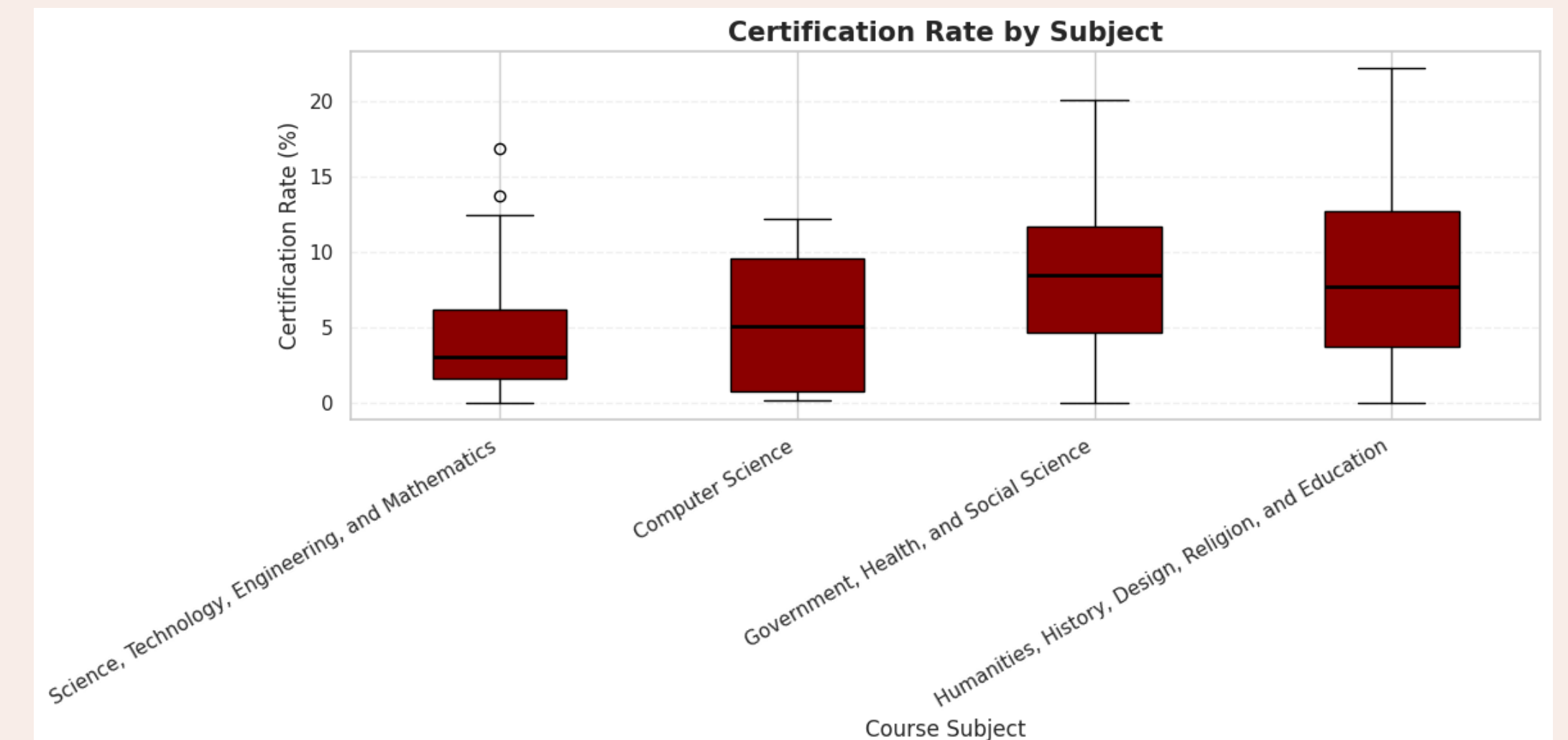
- H_0 : No significant difference in certification rates across different course subject areas
- H_1 : There is a certification rates differ significantly across subject areas .

Methodology



Data Source & Dataset Description:

- HarvardX & MITx dataset (2012–2016)
- 290 courses, 4.5M learners, 250K certifications
- Includes engagement and course data



Conclusion

- Certification rates differ significantly across subjects.
- Some subjects achieve higher completion rates than others.
- MIT and Harvard courses show differences in certification outcomes.
- Most courses have relatively low completion rates.
- Visualizations show variability in certification rates within subjects.

Implications:

- Improving course structure may increase completion rates
- Higher learner engagement can improve certification outcomes